

HYBRID FDMA / TDMA

Channel Allocation & Signaling
Overhead

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25 Carriers × 8 Time Slots

15% reserved for signaling

Net traffic channels = 170

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1. What is Hybrid FDMA/TDMA?

Combining frequency division with time division

FDMA

- Spectrum is divided into frequency carriers.
- Each carrier provides an independent frequency channel.
- Users can be separated by frequency.

TDMA

- Each carrier is divided into time slots.
- Multiple users share the same carrier at different times.
- Users are separated by time.

Hybrid FDMA/TDMA = frequency + time resource sharing

2. Given Parameters

Inputs for the channel-capacity calculation

Number of carriers

25

Time slots / carrier

8

Signaling reservation

15%

Traffic fraction

85%

Goal: determine the net number of slots available for actual user traffic.

3. Total Channel Capacity

Every carrier provides 8 time-slot positions

Total Slots = Number of Carriers × Slots per Carrier

= 25 × 8

= 200 total slots

Before signaling overhead is removed

25 frequency carriers → 8 time slots each → 200 total resource slots

4. Signaling Overhead

15% of all slots are reserved for control information

$$\text{Signaling Slots} = 15\% \times 200$$

$$= 0.15 \times 200$$

$$= 30 \text{ slots}$$

These slots carry network control/signaling rather than normal user traffic.

5. Net Number of Traffic Channels

Removing the reserved signaling capacity

Total slots

200

—

Signaling slots

30

=

170 TRAFFIC CHANNELS

Equivalent calculation: $200 \times (1 - 0.15) = 170$

6. Why Does Signaling Overhead Exist?

Hybrid access still needs control and coordination

- Call/session setup and release
- Channel and time-slot assignment
- Authentication and security
- Synchronization and timing
- Handover and mobility management
- Error, status, and control messages

Signaling tells the network who gets which resources, when, and under what conditions.

7. Why Signaling is Important in Hybrid FDMA/TDMA

Frequency

Choose carrier

Time

Choose slot

Control

Manage users

Mobility

Support handover

Hybrid systems coordinate TWO dimensions

Frequency resources + Time resources + Network control

8. Comparison with Pure FDMA and Pure TDMA

Same percentage overhead does not mean identical control mechanisms

System	Resource division	15% overhead (if specified)
Pure FDMA	Frequency only	30 of 200 = 15%
Pure TDMA	Time only	30 of 200 = 15%
Hybrid FDMA/TDMA	Frequency + Time	30 of 200 = 15%

For this problem, the proportional overhead is 15% in every case; the signaling tasks differ by architecture.

9. Conclusion

$25 \times 8 = 200$ total slots

15% signaling = 30 slots

$200 - 30 = 170$

NET TRAFFIC CHANNELS = 170

Signaling is essential for allocation, synchronization, control, security, and mobility.

Thank You